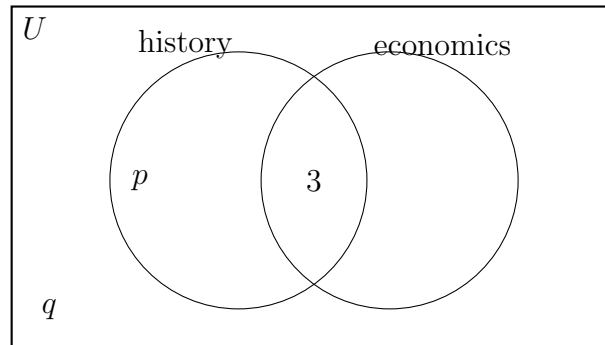


Do Now: Probability exam problems

1. In a group of 20 girls, 13 take history and 8 take economics. Three girls take both history and economics, as shown in the following Venn diagram. The values p and q represent numbers of girls.



(a) Find the value of p ; *[2 marks]*

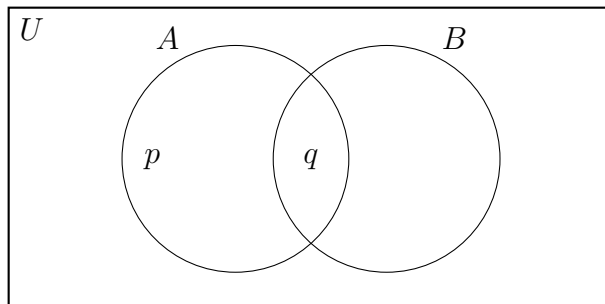
(b) Find the value of q . *[2 marks]*

(c) A girl is selected at random. Find the probability that she takes economics but not history. *[2 marks]*

Working:

(Total 6 marks)

2. The following Venn diagram shows the events A and B , where $P(A) = 0.4$, $P(A \cup B) = 0.8$ and $P(A \cap B) = 0.1$. The values p and q are probabilities.



- (a) (i) Write down the value of q .
- (ii) Find the value of p . *[3 marks]*
- (b) Find $P(B)$. *[3 marks]*
3. Events A and B are independent with $P(A \cap B) = 0.2$ and $P(A' \cap B) = 0.6$.
- (a) Find $P(B)$. *[2 marks]*
- (b) Find $P(A \cup B)$. *[4 marks]*
4. Nene and Dekka both play netball. The probability that Nene will score a goal on her first attempt is 0.75. The probability that Dekka will score a goal on her first attempt is 0.82.
- Calculate the probability that
- (a) Nene and Dekka will both score a goal on their first attempts;
- (b) neither Nene nor Dekka will score a goal on their first attempts. *[4 marks]*